RB 48 Glicko proof of concept:

Proof of Concept: Glicko as a Player-Strength Estimator

Experimental setup

To evaluate whether the Glicko-based model can learn player strength from match results, I created a controlled simulation in which each player is assigned a fixed **true strength**. Teams are then formed from these players and play matches against each other. Match outcomes are generated probabilistically from the teams' underlying strengths, providing a known ground truth against which the evolving Glicko ratings can be compared.

The simulation was run for **1,500 matches**, with the model updating its ratings after every match. At regular intervals, the current Glicko ratings were compared with the players' true strengths. The analysis focuses on five measures: **strength correlation**, **probability correlation**, **probability MSE**, **probability bias**, and **Brier skill relative to baseline predictions**.

An important aspect of the experiment is that the simulated match-outcome function was deliberately made more realistic than a purely linear relationship between strength difference and win probability. Consequently, the numerical scale of the Glicko ratings does **not** have to converge to the numerical scale of the assigned true strengths. The primary question is therefore whether Glicko can **correctly identify the relative strength of players and translate this into increasingly accurate predictions**, rather than whether its rating numbers reproduce the arbitrary true-strength values themselves.

Conclusion

The purpose of this simulation was to test a fundamental assumption behind the RB48 Glicko system: **can a rating system recover the relative strength of players from match results within a reasonable number of games?**

To test this, simulated players were assigned a hidden “true strength” and played matches against one another. The Glicko system had no access to this underlying value and had to infer player strength exclusively from match outcomes. The resulting ratings were then compared with the known ground truth.

The results provide strong evidence that the approach works. The correlation between Glicko rating and true player strength rises rapidly, reaching **0.44 after only 25 games**, **0.75 after 141 games**, and **0.90 after 480 games**. The corresponding Spearman rank correlation shows a similar pattern. This indicates that the system is able to recover the **relative ordering of players** quite quickly, which is the primary purpose of the rating system.

The model also produces increasingly useful win probabilities. Probability correlation rises from **0.43 at 25 games** to **0.77 at 500 games** and **0.89 at 1,500 games**. At the same time, the Brier score consistently outperforms both a 50/50 prediction and the empirical base-rate baseline. This demonstrates that the ratings are not merely producing a plausible ranking: they also contain meaningful predictive information about match outcomes.

Importantly, calibration remains good while predictive resolution improves. The probability bias decreases towards zero, while the Brier decomposition shows low reliability error and increasing resolution. By 1,500 simulated games, the mean predicted win probability differs from the true mean by less than 0.002.

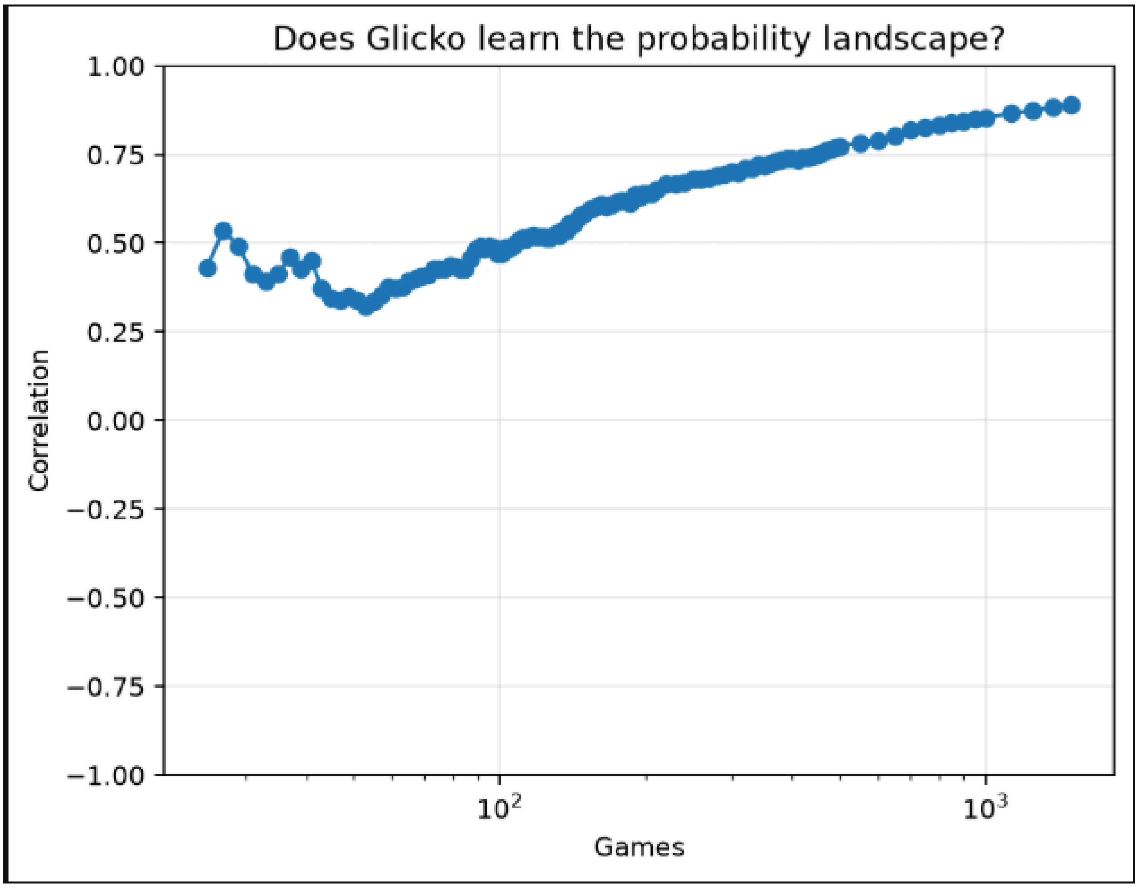
The absolute numerical relationship between Glicko rating and the simulator's “true strength” should not, however, be interpreted as a failure of the model. The match-outcome function used in the simulation determines how differences in underlying strength translate into win probabilities. Changing this function changes the appropriate numerical mapping between rating and strength. Consequently, the **relative ordering and predictive performance are more important than whether Glicko reproduces the simulator's strength values on the same numerical scale**.

Conclusion

This simulation therefore provides the intended proof of concept: **Glicko can reliably recover player rankings from match results within a reasonable amount of data, while simultaneously producing increasingly accurate and well-calibrated predictions.** Further parameter tuning may improve convergence speed and calibration, but the fundamental approach appears sound.

All game counts on the upcoming graphs mean total games, not games per player

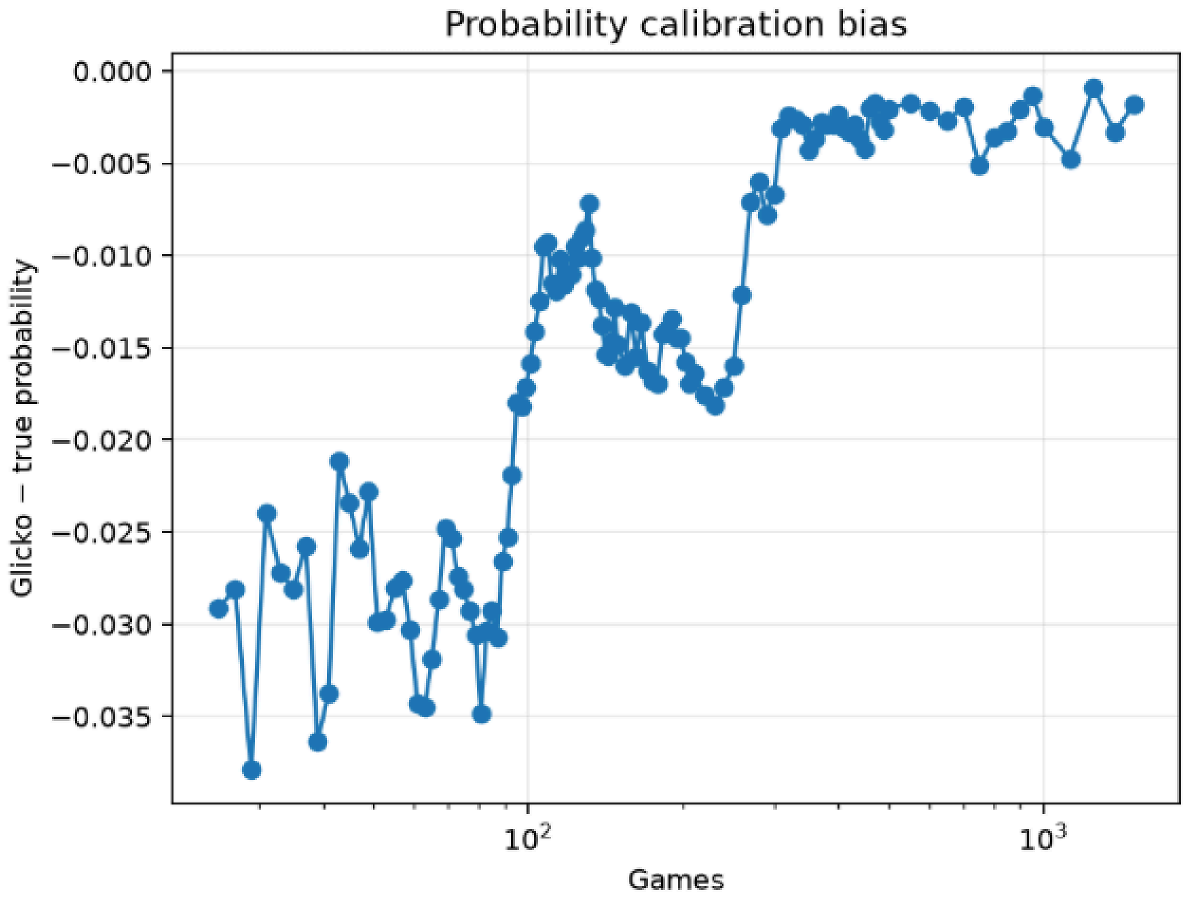
Probability Correlation



What it measures: This is the Pearson correlation between Glicko's predicted win probabilities and the probabilities generated by the simulation's underlying strength model.

Interpretation: The correlation increases from approximately **0.43 after 25 games to 0.89 after 1,500 games**. A correlation of 0.43 indicates a **moderate relationship** between Glicko's predicted probabilities and the simulation's true probabilities: the model is already capturing some of the underlying differences in match strength, but with considerable noise. At 0.89, the relationship is **very strong**, meaning that Glicko's probabilities closely follow the pattern of the true probabilities across matches. The model therefore becomes substantially better at distinguishing relatively even matches from matches with a clear favourite as more results accumulate.

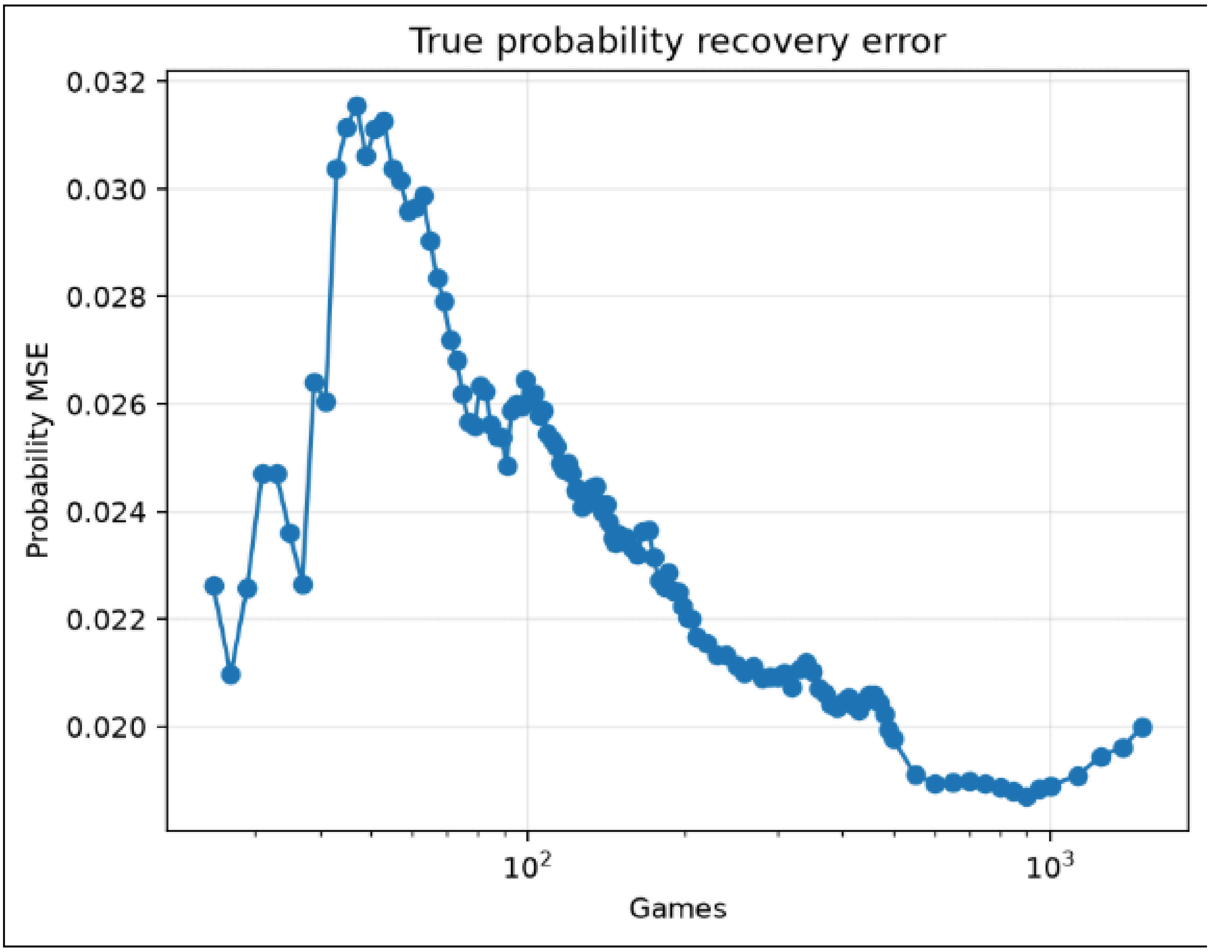
Probability Bias



What it measures: Probability bias is the difference between the average probability predicted by Glicko and the average true probability. A value close to zero indicates that the model does not systematically over- or under-estimate winning chances.

Interpretation: The average bias falls from approximately **−2.9 percentage points early in the simulation to −0.18 percentage points after 1,500 games**. In other words, early predictions slightly underestimate winning probabilities on average, while after sufficient data the systematic error becomes essentially negligible.

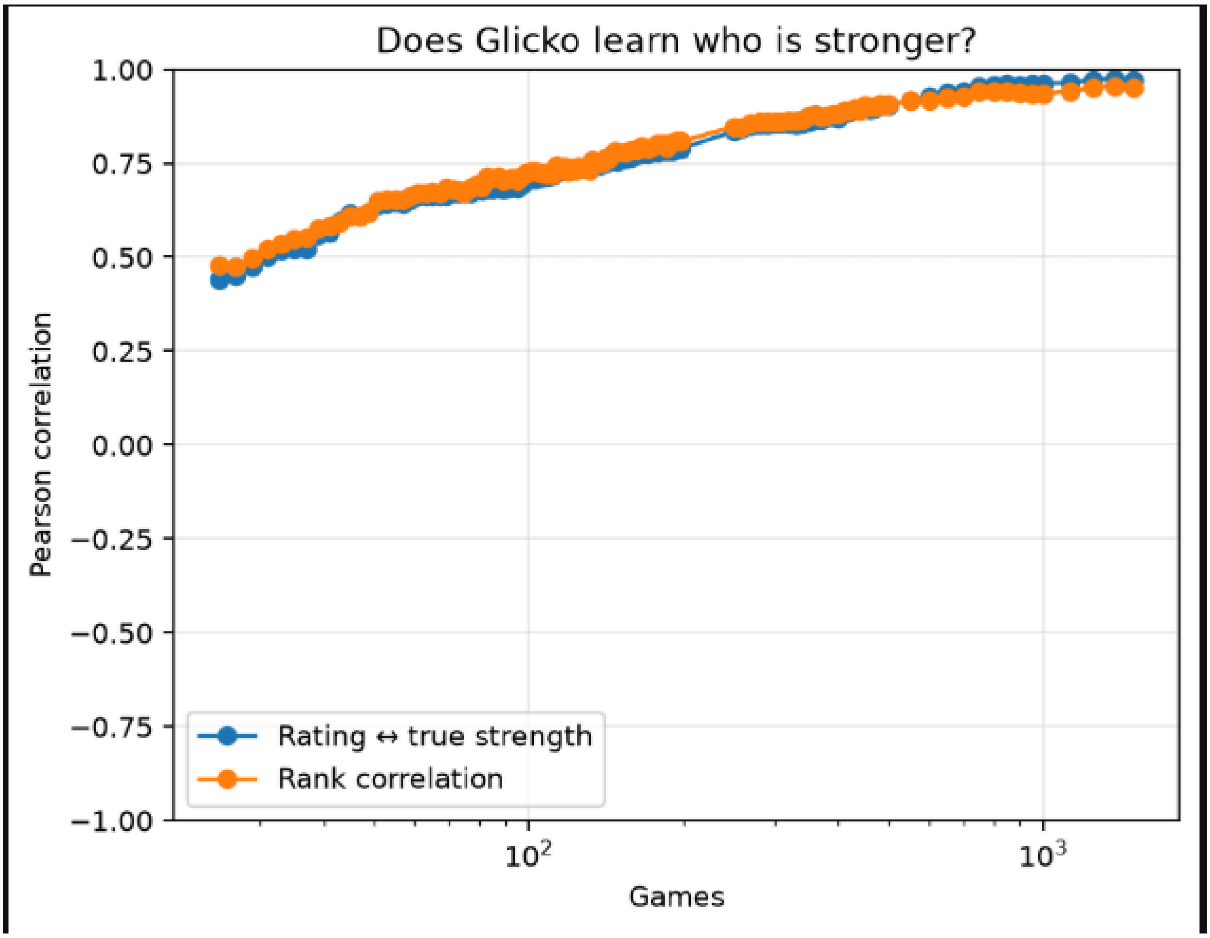
Probability MSE



What it measures: Mean squared error (MSE) measures the average squared difference between Glicko's predicted probability and the simulation's true probability. Lower values indicate closer numerical agreement.

Interpretation: MSE falls from approximately **0.023 to 0.020**. Because this measures the squared difference between predicted and true probabilities, **lower is better**. At 1,500 games, an MSE of 0.020 corresponds to a typical absolute probability error (MAE) of about **0.12**, meaning that predictions are, on average, roughly **12 percentage points away** from the simulation's true probability. The relatively modest change in MSE also illustrates that the model's probability *structure* improves more strongly than its exact numerical scale.

Strength Correlation:



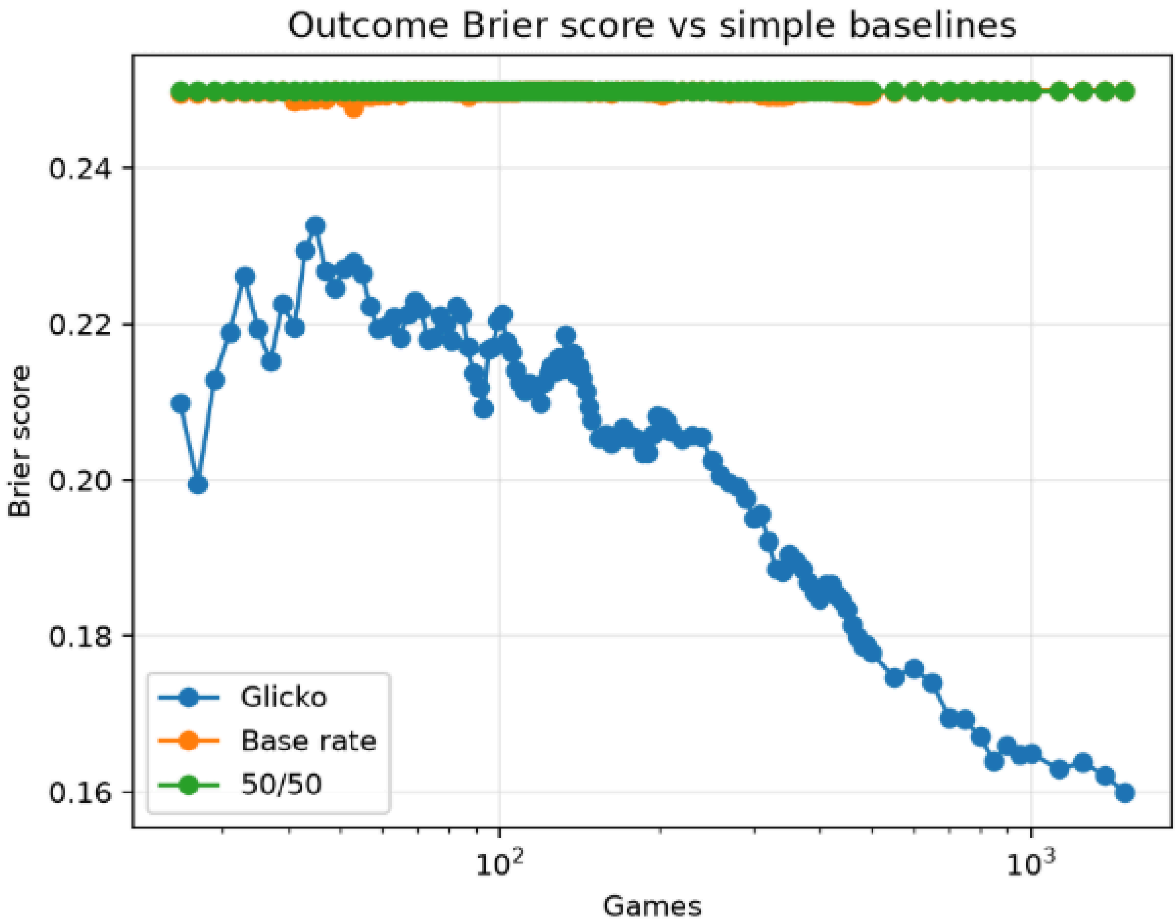
What it measures: This graph compares Glicko ratings with the hidden player strengths used to generate the simulated matches. Pearson correlation measures how closely the numerical ratings track strength, while rank correlation evaluates whether players are ordered correctly.

Interpretation: The correlation rises rapidly: approximately **0.44 after 25 games, 0.75 after 141 games, and 0.90 after 480 games**. This provides strong evidence that Glicko can recover the relative strength hierarchy from match results alone.

A correlation of **0.90 represents a very strong linear relationship** between Glicko rating and the simulated true strength. This means that players with higher underlying strength almost invariably receive higher ratings, and that differences in rating increasingly reflect differences in underlying strength. Importantly, this does **not** mean that the numerical values are identical: Glicko and true strength are on different scales.

A rank correlation of **0.75 indicates a strong correspondence between the Glicko ranking and the true player ranking**. The model has therefore recovered most of the underlying strength hierarchy, although some players are still assigned an incorrect relative position. This happens already after ~150 total games

Brier vs Baseline



What it measures: The Brier score measures the accuracy of probabilistic predictions, with lower values being better. The graph shows the model's Brier skill relative to simple baselines such as predicting 50/50 or the overall base rate.

Interpretation: Positive skill means that Glicko extracts predictive information from the match history that the baseline cannot capture. The steadily improving score indicates that the model becomes increasingly useful as more games are observed.

Data qualification: These results come from simulated matches where the underlying player strengths are known. They demonstrate predictive capability in the controlled simulation, rather than guaranteeing the same performance on real-world match data.